

TRAN HUYNH CHI NGUYEN

AI Engineer Intern

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CAREER OBJECTIVE

Final-year Artificial Intelligence student at HCMUTE with a solid background in Machine Learning, Large Language Models (LLMs), RAG architectures, and MLOps. Hands-on experience developing end-to-end AI applications, from automated crawling and model optimization to scalable multi-container deployment with vLLM and Docker. Seeking an AI Engineer Intern role to apply model optimization and LLM serving techniques to real-world products.

EDUCATION

Ho Chi Minh City University of Technology and Engineering (HCMUTE) (2023 - Expected Graduation: 2027)

- **Major:** Information Technology (Artificial Intelligence)
- **GPA:** 3.1 / 4.0
- **English:** TOEIC 610

TECHNICAL SKILLS

- **Programming:** Python
- **Data & Machine Learning:** Pandas, NumPy, Scikit-learn, PyTorch
- **LLMs & RAG:** LangChain, ChromaDB, vLLM
- **Backend & Storage:** FastAPI, PostgreSQL
- **Tools & Platforms:** Docker, Docker Compose, Apache Airflow, Git, Prometheus, Grafana

FEATURED PROJECTS

1. House Price Forecasting & Recommendation Chatbot System (03/2026 – 09/2026)

Role: Solo Developer | *Source Code:* <https://github.com/nguyentrphanhuynhchi/aura-realestate>

- **Real-world Problem Solved:** Automated price forecasting and demand-driven house recommendation for "Private House" listings in Ho Chi Minh City, helping buyers and agents eliminate hours of manual filtering and providing an objective pricing foundation based on real market data.
- **Automated Data Pipeline:** Engineered an end-to-end data pipeline using `curl_cffi` and `BeautifulSoup4` to collect and clean 5,000+ listings from `Batdongsan.com.vn`, automated via Apache Airflow DAGs with conditional branching and alert callbacks.
- **Machine Learning Valuation Engine:** Built a Stacking Regressor (Base: Ridge, Decision Tree, Random Forest; Meta: Linear Regression) with 5-fold cross-validation and feature engineering (Log-transform, VIF analysis), achieving an R^2 score of 0.7725 on the Test set.
- **Hybrid Search RAG & Conversational Memory:** Architected a Hybrid Search pipeline (ChromaDB all-MiniLM-L6-v2 + BM25 via Reciprocal Rank Fusion) achieving a 94.0% Retrieval Hit Rate @ 2 on a 50-sample ground-truth test set; engineered a 2-tier PostgreSQL JSONB memory managed asynchronously via FastAPI BackgroundTasks.
- **High-Throughput LLM Serving & Streaming:** Deployed Qwen2.5-3B-Instruct (AWQ Marlin W4A16) via vLLM on a 4GB VRAM GPU, sustaining 16 concurrent requests (~600 tokens/request across RAG context & generation) at 95% KV Cache saturation with 0% OOM; integrated Server-Sent Events (SSE) streaming into FastAPI, cutting Perceived TTFT by 80% (~2.3s) with TPOT ~60ms/token.
- **Multi-Container Infrastructure:** Containerized the complete 9-container production ecosystem (FastAPI, React, ChromaDB, PostgreSQL, vLLM, Airflow, Prometheus, Grafana) into isolated networks using Docker Compose.